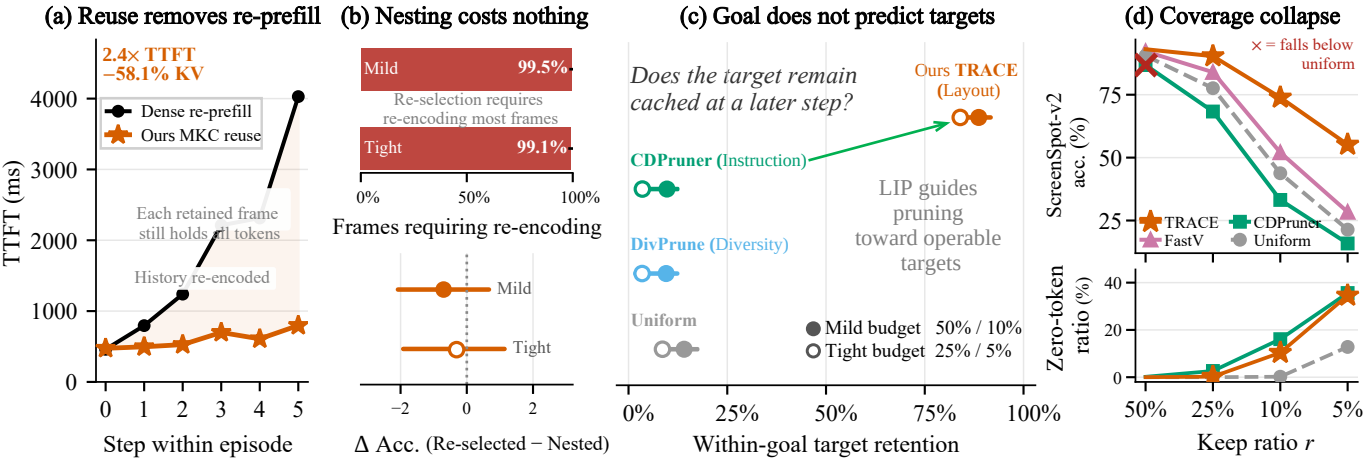

(a) Reuse removes re-prefill
2.4× TTFT
−58.1% KV
Dense re-prefill
Ours MKC reuse
Each retained frame still holds all tokens
History re-encoded
TTFT (ms)
Step within episode
(b) Nesting costs nothing
Mild
99.5%
Re-selection requires re-encoding most frames
Tight
99.1%
Frames requiring re-encoding
Mild
Tight
Δ Acc. (Re-selected − Nested)
(c) Goal does not predict targets
Does the target remain cached at a later step?
Ours TRACE (Layout)
CDPruner (Instruction)
DivPrune (Diversity)
Uniform
LIP guides pruning toward operable targets
Mild budget 50% / 10%
Tight budget 25% / 5%
Within-goal target retention
(d) Coverage collapse
× = falls below uniform
ScreenSpot-v2 acc. (%)
TRACE
FastV
CDPruner
Uniform
Zero-token ratio (%)
Keep ratio r